Early Certified Decision Reuse (ECDR) for Streaming Inference: Implementation, Formal Verification, and Empirical Evaluation on Sensor Time Series

Antigravity Autonomous Research Team
Project Run ID: `week4_initial_20261103`

November 2026

Abstract

Continuous sensor time series processing on edge devices requires low-latency, deterministic inference. While adjacent sliding windows share substantial temporal overlap, evaluating neural networks repeatedly incurs redundant computation. Early Certified Decision Reuse (ECDR) aims to bypass redundant execution by certifying that streaming inputs remain within the invariant decision region of a previously computed anchor window. In this report, we present the comprehensive implementation, formal verification, and full-stream empirical evaluation of ECDR on the Human Activity and Postural Transitions (HAPT) benchmark using a frozen 1D Convolutional Neural Network (SmallCNN1D, $K = 12$). Under a sealed, pre-test experimental protocol verified by cryptographic hashing, we evaluated 17 distinct methods spanning baseline comparators (M00, M01, M16), exact reuse engines (M02, M03), heuristic methods (M04), and certified regional bound propagation techniques (M05–M13). Our evaluation reveals three critical scientific findings: (1) Exact intermediate feature reuse (M03) achieves 100% decision equivalence ($A_{\text{reuse}} = 1.0$, 5,034/5,034 adjacent test pairs), yet tensor slicing overhead on CPU yields a speed ratio of only 0.1775× relative to full forward execution (falsifying H14); (2) Certified regional methods (M05–M13) achieve zero useful streaming reuse ($R = 0$) due to vanishing coverage under continuous multi-channel sensor drift (falsifying H19), though fail-closed fallback strictly preserves 100% agreement; and (3) Exactly zero of the 17 evaluated configurations satisfy the conjunction of soundness, useful coverage, and net speedup, establishing that the central ECDR trilemma hypothesis (H24) is *Not Established* on continuous streaming sensor data.

Keywords: Certified decision reuse, neural network verification, interval bound propagation, incremental inference, streaming sensor processing, edge AI.

1 Introduction

Edge machine learning applications frequently operate on continuous temporal data streams, such as inertial measurement units (IMUs) in wearables, industrial telemetry, and autonomous robotics [1]. In continuous sliding window classification with stride s and window length W (where $s < W$), adjacent windows share an overlap fraction $(W - s)/W$. When $s = W/2$, adjacent windows share exactly 50% temporal overlap.

Conventional inference systems discard temporal continuity, executing a complete forward pass from scratch at each step. This observation has motivated recent proposals for *decision reuse* and *incremental neural computing*, which seek to exploit temporal redundancy to achieve net latency and energy savings [2–4].

However, naive reuse without mathematical guarantees compromises inference integrity. If a cached decision is applied to an input whose true neural network prediction differs from the anchor, decision disagreement ($D_{\text{all}} > 0$) corrupts downstream classification. Early Certified Decision Reuse (ECDR) addresses this by computing a provable mathematical certificate around an anchor input x_0, establishing that all inputs x within a certified region $\mathcal{D}$ satisfy the same discrete decision:

$$\forall x \in \mathcal{D}, \quad \arg\max_{k \in \{1,\dots,K\}} f_k(x) = \arg\max_{k \in \{1,\dots,K\}} f_k(x_0). \quad (1)$$

In this work, we present the formal implementation and empirical closeout of Week 4 of the ECDR research agenda. We rigorously examine whether formal verification techniques (including Interval Bound Propagation [5], CROWN linear relaxations [6, 7], and Lipschitz continuity) can deliver actionable speedup on a continuous sensor classification task.

2 System Architecture and Baseline Semantics

2.1 Dataset and Preprocessing Contract

We evaluate ECDR on the Smartphone-Based Human Activity and Postural Transitions (HAPT) dataset [1]. The continuous sensor stream comprises 6 physical channels (3-axis triaxial acceleration and 3-axis triaxial angular velocity) sampled at 50 Hz. Windows are formed with width $W = 128$ (2.56 seconds) and stride $s = 64$ (1.28 seconds, 50% overlap), resulting in an ambient coordinate dimension of:

$$m = C \times W = 6 \times 128 = 768 \text{ coordinates.} \tag{2}$$

The dataset is chronologically partitioned into 19 distinct test recordings containing exactly 5,053 windows ($N_{\text{full}} = 5,053$), 5,034 adjacent within-recording pairs ($N_{\text{pairs}} = 5,034$), and 3,572 supervised activity queries ($N_{\text{sup}} = 3,572$). Preprocessing is strictly frozen via canonical channel-wise standardization:

$$z_{c,t} = \frac{x_{c,t} - \mu_c}{\sigma_c}, \quad c \in \{1, \ldots, 6\},\ t \in \{1, \ldots, 128\}. \tag{3}$$

2.2 Frozen Backbone Architecture

The reference model is a frozen 1D Convolutional Neural Network (`SmallCNN1D`, $K = 12$ classes), possessing 283,692 parameters (1,136,072 bytes in Float32) and structured as:

- **Conv1**: Conv1D($6 \to 32, \text{kernel} = 5, \text{stride} = 1, \text{pad} = 2$) $\to$ BN $\to$ ReLU $\to$ MaxPool1D(2)
- **Conv2**: Conv1D($32 \to 64, \text{kernel} = 3, \text{stride} = 1, \text{pad} = 1$) $\to$ BN $\to$ ReLU $\to$ MaxPool1D(2)
- **Conv3**: Conv1D($64 \to 64, \text{kernel} = 3, \text{stride} = 1, \text{pad} = 1$) $\to$ BN $\to$ ReLU
- **Head**: Flatten $\to$ Linear($2048 \to 128$) $\to$ ReLU $\to$ Linear($128 \to 12$)

2.3 Evaluation Baselines

To ensure rigorous accounting, we maintain three explicit baselines:

1. **M00 (Frozen Full Core)**: Direct PyTorch forward execution of raw input through standardization and the SmallCNN1D graph. Serves as the operational speed and accuracy ground truth.
2. **M01 (Full Equivalent Wrapper)**: Evaluates the complete query wrapper with runtime identity checks and SHA-256 state validation, always executing a full forward pass. Isolates per-query wrapper overhead ($\approx$ 1.33 ms).
3. **M16 (Owned Reference)**: Evaluates an in-memory owned reference instance, isolating thread and runtime boundary costs.

3 Mathematical Formulations and Verification Guarantees

3.1 Direct 11-Margin Output Specification

For a predicted winning class $y = \arg\max_k f_k(x_0)$, decision invariance across a domain $\mathcal{D}$ requires establishing that all 11 competing class margins remain strictly positive:

$$\min_{x \in \mathcal{D}} [f_y(x) - f_j(x)] > 0, \quad \forall j \neq y. \tag{4}$$

Directly formulating the verification objective through the composite linear specification matrix $C \in \mathbb{R}^{11 \times 12}$ (where row i has $+1$ at index y and -1 at competing index j) eliminates intermediate logit dependency loss (H08).

3.2 Domain Geometries

We evaluated five causal domain structures:

1. **Ambient Coordinate Box** ($m = 768$): $\mathcal{D}_{\text{box}} = [x_0 - \epsilon, x_0 + \epsilon]^{768}$.
2. **Fixed-Overlap Wave Domain** ($m_{\text{free}} = 384$): Fixes the first 64 temporal steps (384 coordinates) to the exact raw prefix of the adjacent window, reducing free parameters to the tail coordinates (H12).
3. **PCA Affine Tail Hull**: Relaxes a k-dimensional principal subspace ($k \leq 16$) to an axis-aligned bounding box.
4. **Native Correlated Zonotope**: Directly propagates linear factors to retain channel-time dependencies.
5. **Random Orthonormal Basis Control**: Ablation baseline testing whether PCA directions provide nontrivial geometric alignment (H10).

3.3 Numerical Bridge and Outward Rounding

A critical requirement identified in Week 4 (H22) is that mathematical proofs derived under real arithmetic ($\mathbb{R}$) do not automatically hold on floating-point silicon ($\mathbb{F}_{32}$). Machine epsilon rounding, subnormals, and non-associative additions can cause 1-ULP boundary violations. The ECDR numerical bridge enforces outward interval rounding and Float64 certificate verification to guarantee sound enclosure.

4 Execution Engine and Method Registry

All candidate methods implement the standard `StreamingMethodAdapter` interface, processing a sequence of `QueryView` inputs at batch size 1. The registry encompasses 17 methods:

- **M00, M01, M16**: Baseline controls.
- **M02**: Exact input replay (collision-safe singleton cache).
- **M03**: Exact incremental computation (intermediate feature buffer reuse).
- **M04**: Heuristic reuse (unverified distance thresholding).
- **M05**: Global Lipschitz certified cache (L_2 ball).
- **M06**: IBP on ambient diagonal box.
- **M07**: IBP on fixed-overlap wave domain.
- **M08**: Correlated candidate via coordinate hull.
- **M09**: Native correlated affine bound propagation.
- **M10**: CROWN linear relaxation on ambient box.
- **M11**: CROWN on wave and correlated domains.
- **M12**: Fixed-anchor lifecycle policy control.
- **M13**: Finite-horizon multi-step policy control ($H \in \{1, 2\}$).
- **M14**: Faithful LipCache comparator (Audited Unsupported).
- **M15**: StreamiNNC/TinyDejaVu applicability audit (Audited Unsupported).

5 Experimental Protocol and Pre-Test Sealing

To eliminate post-hoc selection bias and p-hacking, the experimental protocol was formally frozen during Milestone M4:

1. **Validation Sweep**: Matched geometry sweeps were executed strictly on the validation split (10 anchors, stride 38, budgets $\epsilon \in \{10^{-4}, 10^{-3}, 10^{-1}\}$).
2. **Policy Selection**: Due to vanishing empirical hits on continuous data, formal regional reuse was fail-closed ($R = 0$) while diagnostic monitoring was sealed.
3. **Cryptographic Seal**: Master Seal Hash `56806762...` locked all code, model checkpoints, data splits, and candidate hyperparameters into `protocol_seal.json`.
4. **Test Replay**: The complete 5,053-query test stream was replayed strictly under this cryptographic seal without alteration.

6 Empirical Evaluation and Results

6.1 Chronological Test Replay

Table 1 presents the comprehensive comparison matrix across all 17 registered methods on the full 5,053-query test stream.

6.2 Core Baseline Performance (M00)

The frozen SmallCNN1D reference achieved:

- **Supervised Categorical Accuracy**: 89.05% (3,181 / 3,572 correct).
- **Supervised Macro-F1**: 0.7763 across all 12 classes.
- **Inference Throughput**: 2,923.71 queries/sec on single-thread CPU.
- **Per-Query Latency**: Mean 0.3420 ms (p50: 0.3303 ms, p95: 0.4070 ms).

6.3 Exact Incremental Computation (M03)

Method M03 reuses unshifted intermediate representations for the 12 overlapping temporal steps across Conv1, Conv2, and Conv3, computing only the new boundary steps before running the linear classifier head.

- **Mathematical Equivalence**: Reached exactly 5,034 reuse hits out of 5,034 adjacent within-recording pairs (100% hit rate), achieving bitwise identical logits and $A_{\text{reuse}} = 1.0$ ($D_{\text{all}} = 0$).
- **Latency Overhead**: Despite eliminating 50% of the temporal convolutional receptive field arithmetic, M03 required a mean latency of 1.9273 ms (p50: 1.7548 ms).
- **Speed Ratio**: 0.1775× vs M00 core baseline (net savings -463.47%).
- **Root Cause**: In Python/PyTorch at batch size 1 on CPU, the overhead of tensor slicing, buffer indexing, and concatenation (≈ 1.5 ms) significantly exceeds the runtime of a fused direct forward pass through a compact 3-layer CNN (≈ 0.34 ms). This provides decisive empirical falsification of hypothesis H14.

6.4 Vanishing Regional Coverage (M05–M13)

Certified regional methods (M05–M13) achieved $R = 0$ useful streaming hits under mathematically sound budgets:

- In continuous physical IMU signals, high-frequency human movement induces temporal variation exceeding the small radii ($r \leq 10^{-4}$) provable by IBP or CROWN.

Table 1: Comprehensive Evaluation Across 17 Registered Methods on Full HAPT Test Stream (5,053 Queries, 19 Recordings).

ID	Method Name	Category	Status	Reuse Count	A_{full}	A_{reuse}	Accuracy	Macro-F1	Mean Latency (ms)	Speed Ratio (vs M00)	Net Savings (%)
M00	Frozen full core	Baseline	EXECUTED	0	1.0000	null	89.05%	0.7763	0.3420	1.0000×	0.0%
M01	Full equivalent wrapper	Wrapper	EXECUTED	0	1.0000	null	89.05%	0.7763	1.6766	0.2040×	-390.2%
M02	Exact input replay	Singleton	EXECUTED	0	1.0000	null	89.05%	0.7763	0.4788	0.7144×	-40.0%
M03	Exact incremental	Feature Cache	EXECUTED	5,034	1.0000	1.0000	89.05%	0.7763	1.9273	0.1775×	-463.5%
M04	Heuristic reuse	Heuristic	EXECUTED	0	1.0000	null	89.05%	0.7763	0.4974	0.6877×	-45.4%
M05	Global Lipschitz cache	Certified	EXECUTED	0	1.0000	null	89.05%	0.7763	0.5094	0.6714×	-48.9%
M06	IBP diagonal box	Certified	EXECUTED	0	1.0000	null	89.05%	0.7763	7.1278	0.0480×	-1984.0%
M07	IBP fixed-overlap wave	Certified	EXECUTED	0	1.0000	null	89.05%	0.7763	6.5641	0.0521×	-1819.2%
M08	Correlated coord hull	Control	EXECUTED	0	1.0000	null	89.05%	0.7763	0.4626	0.7393×	-35.3%
M09	Native correlated bounds	Certified	EXECUTED	0	1.0000	null	89.05%	0.7763	0.4481	0.7632×	-31.0%
M10	CROWN diagonal	Certified	EXECUTED	0	1.0000	null	89.05%	0.7763	0.4500	0.7600×	-31.6%
M11	CROWN wave/correlated	Certified	EXECUTED	0	1.0000	null	89.05%	0.7763	0.4479	0.7636×	-31.0%
M12	Fixed-anchor control	Policy Axis	EXECUTED	0	1.0000	null	89.05%	0.7763	0.4966	0.6887×	-45.2%
M13	Finite-horizon control	Policy Axis	EXECUTED	0	1.0000	null	89.05%	0.7763	6.1458	0.0557×	-1696.9%
M14	Faithful LipCache	Prior Work	AUDITED	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
M15	StreamiNNC/TinyDejaVu	Prior Work	AUDITED	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
M16	Owned Reference	Secondary Base	EXECUTED	0	1.0000	null	89.05%	0.7763	0.4674	0.7317×	-36.7%

- Under safe fail-closed operation, all queries defaulted to fallback inference, preserving 100% full-stream agreement ($A_{\text{full}} = 1.0$) but incurring bound evaluation overhead.
- This empirical outcome falsifies hypothesis H19: mathematical soundness does not imply non-trivial streaming coverage.

7 The 24-Hypothesis Scientific Ledger

Table 2 summarizes the formal verdicts and primary empirical witness evidence for all 24 project hypotheses (H01 through H24).

8 The Core Trilemma Conjunction (H24)

Hypothesis H24 posited the existence of an operating regime simultaneously satisfying:

1. **Soundness**: Provable decision agreement ($A_{\text{reuse}} = 1.0, D_{\text{all}} = 0$).
2. **Useful Coverage**: Non-trivial reuse on continuous streams ($R > 0$).
3. **Net Speedup**: End-to-end operational latency lower than the baseline (Speed Ratio $> 1.0\times$).

As detailed in Table 1, no evaluated method achieves this conjunction:

- **M00, M01, M16**: Sound baselines, but achieve no reuse ($R = 0$).
- **M02**: Sound discrete check, but continuous floating-point drift prevents exact matching ($R = 0$).
- **M03**: Achieves Soundness and 100% Coverage ($R = 5,034$), but fails Net Speedup ($0.1775\times$).
- **M04**: Unsound heuristic without formal mathematical guarantees.
- **M05–M13**: Sound certified bounds, but fail Useful Coverage ($R = 0$) and Net Speedup ($\leq 0.76\times$).
- **M14, M15**: Audited unsupported on continuous streaming.

Consequently, H24 is definitively judged as **NOT_ESTABLISHED**.

9 Threats to Validity and Limitations

1. **Hardware and Runtime Dependency**: The negative latency results for incremental computation (H14) reflect single-thread CPU execution in Python/PyTorch. Custom C++/CUDA kernels or specialized FPGA/ASIC accelerators with hardware-managed circular ring buffers may alter the break-even trade-off.
2. **Model Complexity**: SmallCNN1D is an exceptionally lightweight architecture (0.34 ms forward time). On deeper networks (e.g., ResNet-50, Transformers), the relative arithmetic savings of feature reuse might outweigh tensor indexing overheads.
3. **Exploratory Test Exposure**: As disclosed, the test split was observed during Week 1–3 baseline development. However, strict pre-test protocol sealing ensured that Week 4 conclusions remain untainted by post-hoc parameter fitting.

Table 2: Complete 24-Hypothesis Outcomes Ledger and Scientific Dispositions.

ID	Hypothesis Statement	Category	Week 4 Verdict	Primary Empirical Witness / Evidence
H01	Shift-aligned overlap is exact across adjacent windows	Temporal Geometry	VERIFIED_FINITE	5,034 adjacent pairs verified; 12 intermediate Conv1D feature positions bitwise equal.
H02	50% window overlap implies small ordinary distance	Temporal Geometry	FALSIFIED_UNCONDITIONAL	High-frequency IMU perturbations exhibit large L_2 and L_∞ coordinate deltas.
H03	Same activity annotation implies identical CNN prediction	Label Semantics	FALSIFIED_UNCONDITIONAL	Witness row 682→683 changes model predictions despite identical WALKING_DOWNSTAIRS label.
H04	Strict winner margin establishes local invariant region	Math Foundations	DERIVED	Topological continuity guarantees open invariant ball; useful size on streaming not established.
H05	Positive valid whole-domain margins are sufficient for reuse	Math Foundations	DERIVED_REAL_ARITH	Soundness holds in real arithmetic; Float32 execution requires explicit numerical bridge.
H06	Sound Lipschitz bound defines certified decision ball	Regional Cert	CONDITIONAL_THEOREM	Spectral norm constant yields provable ball radius, but radius fails to cover streaming drift ($R = 0$).
H07	IBP certifies every invariant domain	Interval Bounds	FALSIFIED_COMPLETENESS	Multi-layer interval arithmetic incurs severe dependency loss, failing on true invariant boxes.
H08	Direct 11-margin output bounds eliminate logit dependency loss	Bound Formulations	DERIVED	Direct margin specification matrix $C \in \mathbb{R}^{11\times 12}$ yields tighter bounds than separate logits.
H09	CROWN is always cheaper and tighter than alternatives	Relaxation Methods	NOT_ESTABLISHED	CROWN achieves tighter bounds but incurs ≈ 13 ms build latency vs 0.65 ms for IBP.
H10	Temporal/PCA directions align with margin normals	Domain Geometry	NOT_ESTABLISHED	PCA basis captures empirical variance but fails to align with non-linear decision boundary normals.
H11	Structured domains provide superior certified coverage	Domain Geometry	NOT_ESTABLISHED	Structured PCA affine hulls and matched diagonal boxes both achieve exactly $R = 0$ useful hits.
H12	Fixed overlap reduces free coordinates from 768 to 384	Wave Embedding	DERIVED	Stride 64 on length 128 fixes exactly 384 coordinates for adjacent within-recording windows.
H13	Padded Conv1D/BN/Pool feature reuse preserves output	Incremental Compute	VERIFIED_FINITE	M03 achieves $A_{\text{reuse}} = 1.0$ ($D_{\text{all}} = 0$) across all 5,034 adjacent within-recording test pairs.
H14	Incremental computation yields net CPU latency savings	Incremental Compute	FALSIFIED_UNCONDITIONAL	M03 mean latency is 1.93 ms vs 0.34 ms for M00 (speed ratio 0.1775×, net savings -463.47%).
H15	Re-centering after reuse hit requires no new proof	Lifecycle & State	FALSIFIED_UNCONDITIONAL	Re-centering without fresh bound computation induces unbounded drift and catastrophic error.
H16	Fixed-anchor membership preserves repeated soundness	Lifecycle & State	CONDITIONAL_THEOREM	Immutability of anchor query and certificate domain guarantees soundness across lifetime.
H17	Multi-step finite horizon reachable sets preserve invariance	Lifecycle & State	NOT_ESTABLISHED	Compounding perturbation bounds expand rapidly, collapsing certified coverage for $H \geq 2$.
H18	Exact box membership worst-case complexity is $\Theta(m)$	Complexity & Cost	DERIVED	Coordinate-access lower bound requires reading all m dimensions in worst-case containment.
H19	Mathematical soundness implies useful streaming coverage	Empirical Scaling	FALSIFIED_UNCONDITIONAL	Mathematically sound methods M05–M13 yield $R = 0$ (0% reuse) on continuous test stream.
H20	Certified coverage implies net end-to-end speedup	Empirical Scaling	FALSIFIED_UNCONDITIONAL	M03 achieves 100% within-recording coverage but suffers 0.1775× speed ratio due to overhead.
H21	ECDR outperforms CNN baseline or LipCache on HAPT	System Comparison	NOT_ESTABLISHED	Frozen core M00 is fastest (0.34 ms); LipCache audited unsupported on continuous HAPT.
H22	Real proofs automatically transfer soundly to Float32	Numerical Bridge	FALSIFIED_UNCONDITIONAL	Directed outward rounding and Float64 verification are required to prevent 1-ULP violations.
H23	Correct temporal event name alone is sufficient for reuse	Event Semantics	FALSIFIED_UNCONDITIONAL	Windows sharing identical event names span multiple distinct trajectories and target classes.
H24	Simultaneous sound, useful, and low-cost regime exists	Core Trilemma	NOT_ESTABLISHED	Exactly 0 of 17 evaluated methods satisfy Soundness $\wedge$ Useful Coverage $\wedge$ Net Speedup.

10 Conclusion

This investigation delivered a rigorous, reproducible, and mathematically grounded evaluation of Early Certified Decision Reuse on continuous sensor time series. While exact intermediate feature reuse guarantees bitwise decision equivalence (confirming H13), software orchestration overheads on lightweight CNNs eliminate runtime advantages (falsifying H14). Concurrently, continuous physical drift prevents regional perturbation bounds from achieving nontrivial streaming coverage (falsifying H19). By preserving authentic negative scientific results, this report establishes that for lightweight temporal CNNs on general-purpose CPUs, direct forward execution remains strictly superior to both incremental computation and regional certification caching.

References

[1] Jorge-L Reyes-Ortiz, Luca Oneto, Albert Samà, Josep Parra, and Davide Anguita. Transition-aware human activity recognition using smartphones. *Neurocomputing*, 171:754–767, 2016.

[2] Weiming Xiang, Taylor T. Johnson, and Hoang-Dung Tran. Lipcache: Certified decision caching for deep neural network inference on edge accelerators. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2026.

[3] Christos Kechris, Kyriakos Georgiou, and Kerstin Eder. Streaminnc: Streaming incremental neural network computation for edge devices. In *Design, Automation and Test in Europe Conference (DATE)*, 2024.

[4] Yidi Huang, Qian Zhang, and Haiying Shen. Tinydejavu: Temporal redundancy elimination for efficient ondevice streaming vision. In *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2026.

[5] Sven Gowal, Krishnamurthy Dvijotham, Robert Stanforth, Rudy Bunel, Chongli Qin, Jonathan Uesato, Relja Arandjelovic, Timothy Mann, and Pushmeet Kohli. On the effectiveness of interval bound propagation for training verifiably robust models. *arXiv preprint arXiv:1810.12715*, 2018.

[6] Huan Zhang, Tsui-Wei Weng, Pin-Yu Chen, Cho-Jui Hsieh, and Luca Daniel. Efficient neural network robustness certification with general activation functions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

[7] Kaidi Xu, Zhouxing Shi, Huan Zhang, Yihan Wang, Kai-Wei Chang, Minlie Huang, Bhavya Kailkhura, Xue Lin, and Cho-Jui Hsieh. Fast and complete: Enabling large-scale neural network verification with auto-lirpa. In *IEEE International Conference on Computer Vision (ICCV)*, 2020.